

Madhavan Panneerselvam Kumar

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EDUCATION

DePaul University, Chicago, IL

Dec 2026

M.S. Artificial Intelligence

SRM University, India

Apr 2024

M.S. Applied Data Science

Sri Ramachandra University, India

Apr 2022

B.S. Data Science

SKILLS

Languages: Python (pipelines, data workflows, backend services), SQL, R, C++, Excel (pivot tables, charts, statistical models)

Data Analysis & Modeling: pandas, NumPy, scikit-learn, XGBoost, statistical modeling, hypothesis testing, regression analysis, anomaly detection, forecasting, feature engineering, SHAP, TF-IDF

ML & AI: PyTorch, TensorFlow, LangChain, OpenAI API, RAG architectures (chunking, embeddings, reranking), ChromaDB (vector DB), RLHF, PPO, reward modeling, SFT fine-tuning, transformer architectures, model-based RL (DreamerV1/RSSM), prompt engineering, agentic workflows

Visualization & Reporting: Power BI, Tableau, Excel dashboards, Streamlit — creating reports and dashboards to communicate findings to technical and non-technical audiences

Data Workflow & Infrastructure: ETL pipelines, Alteryx, data collection, validation, cleaning, transformation, AWS, GCP (Vertex AI), Azure (familiar), Git, GPU workload profiling, Streamlit (prototyping)

Communication & Research: requirements gathering, SOP documentation, cross-functional collaboration, stakeholder communication, problem-solving, attention to detail, presentation skills, Microsoft 365 (Word, PowerPoint, Excel, Outlook)

WORK EXPERIENCE

Machine Learning Intern - Global Techno Solutions, India

Jun - Sep 2023

- Built Python ML pipelines (scikit-learn, TensorFlow) with automated preprocessing, model training, and inference; collaborated with data engineers to validate datasets and documented experiments with versioning — cutting pipeline runtime by 30%.

Data Analytics Process Automation Intern - Alteryx, India

Jun - Aug 2022

- Built ETL pipelines in Alteryx automating 5+ manual workflows across analyst teams — cutting processing effort by 60%; documented findings as SOPs and created reports for non-technical stakeholders.

PROJECTS

Agentic Cleantech AI Assistant | Python, *LangChain*, *GPT-4o-mini*, *OpenAI API*, *ChromaDB*, *Tavily Search*, *RAG*

- Designed and built an agentic RAG system connecting an LLM to domain-specific knowledge via modular data connectors. Used LangChain to orchestrate multi-step tool-calling (vector retrieval, live web search, direct generation), ChromaDB as the vector store, and the OpenAI API with structured outputs and retry logic.

- Hit 100% retrieval accuracy on CleanTech-50 by tuning chunk size, overlap, and reranking parameters instead of accepting defaults. Built audit trail logging and access controls so every retrieval was traceable. Encoded domain expertise into agent instructions and behavioral guardrails. Validated with real users iteratively.

RLHF Pipeline on GPT-2 | Python, *PyTorch*, *GPT-2*, *SFT*, *reward modeling*, *PPO*, *trl*, *Hugging Face transformers*

- Implemented a complete RLHF training stack: supervised fine-tuning on Dolly-15k, reward model trained on 160K human preference pairs (Anthropic HH-RLHF dataset), and PPO-based policy optimization using the reward model as the feedback signal.

- Designed evaluation datasets and tracked reward curves + KL divergence across all three training stages. Ran ablations on reward model capacity and KL penalty coefficients to map where training was sensitive. Documented all methodology in a reproducible format.

DreamerV1 World Model - Autonomous Highway Driving | Python, *PyTorch*, *RSSM*, *highway-env*, *Gymnasium*, *model-based DRL*

- Implemented a full DreamerV1 world model from scratch in PyTorch: recurrent state-space model (RSSM) for latent dynamics, image encoder/decoder, and learned reward head. The agent learns entirely in imagination by rolling out trajectories in the learned world model.

- Built parallel rollout collection and experience replay pipelines. Profiled GPU utilization and optimized for memory efficiency. Iterated on reward shaping and observation design to fix degenerate policies. Wrote a ~5,100-word academic report (LaTeX) documenting architecture decisions, ablations, and findings.

AI vs. Human Text Classifier | Python, *scikit-learn*, *TF-IDF*, *Random Forest*, *SVM*, *MLP*, *SHAP*

- Built a classification pipeline across four model architectures achieving 99.82% accuracy on a 500K+ sample dataset. Ran SHAP analysis to verify whether models learned genuine signal or dataset artifacts. Measured inference speed and memory footprint per classifier. Documented full methodology for reproducibility.

F1 Telemetry Prediction Model | Python, *SQL*, *pandas*, *XGBoost*, *scikit-learn*, *feature engineering*, *statistical modeling*

- Built an ML pipeline on F1 race telemetry: ingested high-frequency sensor data, handled noise and missing values, engineered 20+ features from raw signals. Trained models achieving RMSE 6.02s on lap time forecasting and AUC 0.72 on event classification. Compared runtime and accuracy across model architectures.

Image Enhancement for Traffic Surveillance | Python, *OpenCV*, *NumPy*, *image processing*, *edge detection*, *segmentation*

- Designed image processing pipelines using OpenCV and NumPy to perform filtering, edge detection, and segmentation on real-time traffic video data. Improved feature extraction for traffic signal and lane detection. Validated image quality improvements using PSNR and SSIM metrics.

CERTIFICATIONS

Prompt Design in Vertex AI (Google) · Google AI Essentials (Coursera) · Introduction to AI (Google) · Introduction to Artificial Intelligence (IBM) · Data Analytics with Python (NPTEL) · Python for Data Science (NPTEL)